Aim:

To verify Beer's Law using Fe+3 solution by spectrophotometer / colorimeter and to determine the concentration of a given unknown Fe+3 solution

Instrument required: Colorimeter/ Spectrophotometer

Glassware required: Volumetric flask, Pipette

Reagent required:

1. Stock solution of ferrous ion
2. HCL
3. Potassium thiocyanate
4. Acetic acid
5. Conc. nitric acid.

Theory: Chemicals analysis through measurement of absorption of light radiation in visible region (400-700) of spectrum is known as spectrophotometry. A device which measure the percentage transmittance of light radiation when light of certain intensity and frequency range is passed through the sample is known as spectrophotometer. It compares the intensity of the transmitted light with that of the incident radiation.

Transmittance (T) of a solution is the fraction of incident light transmitted by solution mathematically: $T = I_t / I_0$, where I_t is the intensity of transmitted light and I_0 is the intensity of a beam of monochromatic light.

Absorbance (A) or optical density (OD) is defined as $A = \log I_o / I_t$.

From the Lambert – beer's Law $A = \varepsilon Ct$

Where, ε is molar extinction coefficient (or molar absorptivity),

C is concentration (in gms/L) of the absorbing medium,

T is thickness of the absorbing medium.

If same sample cell (t is constant) is used for measurement of absorbance by solutions of different concentration, then $A \propto C$

i.e. the extent of absorbance (A) is directly proportional to the concentration (C) of the absorbing medium.

Thus if a graph is plotted between A and C we get a straight line for solution obeying Lambert-Beer's Law. This is known as calibration curve.

Procedure:

A. Preparation of stock solution (1000 ppm) of iron: Dissolve 4.836g of ferric chloride in 0.1 M HCl and dilute it to 1 litre.

B. Potassium thiocyanate solution: take 10g of potassium thiocyanate and dissolve it in distilled water, adjust the pH with the help of conc. nitric acid (4 ml) and acetic acid (10 ml), make the volume up to 1 litre.

C. Preparation of working standard:

100 ppm working standard solution: Take 10 ml of working stock solution (1000 ppm) and dilute it upto 100 ml by distilled water.

1. 2 ppm=Dilute 2 ml of working stock solution 100 ppm upto 100 ml with distilled water.
2. 4 ppm= Dilute 4 ml of working stock solution 100 ppm upto 100 ml with distilled water.
3. 6 ppm= Dilute 6 ml of working stock solution 100 ppm upto 100 ml with distilled water.
4. 8 ppm= Dilute 8 ml of working stock solution 100 ppm upto 100 ml with distilled water.
5. 10ppm= Dilute 10 ml of working stock solution 100 ppm upto 100 ml with distilled water.

D. Preparation of solution for calibration curve: Take 5 ml working standard (1-5) in 5 test tube, add 1 ml potassium thiocyanate in each test tube.

Observation Table:

Sl.No.	Concentration in ppm	Absorbance
1	2 ppm	
2	4 ppm	
3	6 ppm	
4	8 ppm	
5	10 ppm	
6	Unknown Sample	

Result:

The conc. Of Fe 3+ ion in a given sample of water using colorimeter = ………mg/l

Precaution:

1. Measure absorbance value only at 480 nm. Set absorbance zero for blank.
2. Cuvette is fragile and it should be used with great care.